

in high-level athletes and patients sustaining minor trauma has been associated with anterior labral tears and anterior acetabular chondral defects in many of these patients.^{5,15} Patients with mechanical complaints such as clicking or locking often have loose bodies or labral tears, whereas patients with traumatic or inflammatory conditions tend to have anterior groin pain, stiffness, and limitation of motion.

PHYSICAL EXAMINATION

The physical examination should begin with measurement of temperature and vital signs. Although hip infections are rare in adults without a history of prior hip surgery, hip pyarthrosis should be considered in a febrile patient with hip pain. Other causes of hip discomfort and fever include psoas abscess, prostatitis, pelvic inflammatory disease, and urinary tract infection.

Gait is an important part of the hip evaluation. Trendelenburg gait often is a good indicator of an intraarticular hip disorder, but also is seen in patients with extraarticular problems. Pelvic obliquity suggests a leg-length discrepancy or scoliosis, both potential sources of hip pain. A thorough examination of the lumbar spine, including range of motion (ROM), motor, sensory, and reflex testing of the lower extremities, and straight leg testing can help eliminate lumbar disorders as a cause of hip pain.

The position of the hip at rest can provide some useful information. The hip capsule's largest potential volume is with the hip flexed, abducted, and externally rotated, and patients with synovitis or effusions tend to hold the hip in this position. Trendelenburg's test can be used to evaluate a patient with deficient or painful abductor muscles. The patient stands on one leg with the contralateral hip and knee flexed. Patients with normal abductors will lift the pelvis contralateral to the stance limb. Patients with abductor weakness or pain will allow the pelvis to drop contralateral to the stance leg or shift the upper body over the stance leg to protect the weak abductors.

Examination of the hip should continue with palpation of bony prominences, ROM,

and provocative testing. Pain over the greater trochanter may indicate trochanteric bursitis. Patients with intraarticular or extraarticular hip disorders often will have limited ROM when compared with the normal hip. Provocative tests for labral tears include moving the hip from a flexed, externally rotated, and abducted position to an extended, internally rotated, and adducted position to test for anterior disorders, and moving from a flexed, internally rotated and adducted position to an extended, abducted, and externally rotated position to test for posterior disorders. If positive, these tests produce a painful click or burning sensation.⁴ Anesthetic injections about the hip also can be a useful diagnostic tool in difficult cases.

For completeness, it is important to consider sources of referred pain. The groin should be examined for femoral or inguinal hernias. Osteitis pubis, athletic pubalgia, and adductor tendinitis all can produce groin pain that mimics pain associated with disorders of the hip. An abdominal examination should be done to rule out gastrointestinal sources of pain. Other sources of referred pain include renal problems such as nephrolithiasis, urinary tract infection, and prostatitis, gynecologic problems including ovarian cysts and pelvic inflammatory disease, and urologic problems in males such as epididymitis, urethritis, and hydrocele. Femoral aneurysms and pseudoaneurysms also can present as hip pain and typically are diagnosed as a palpable, pulsatile mass. Other musculoskeletal sources of hip pain include the lumbar spine, the sacroiliac joints, the thigh, and the knee.

LABORATORY EVALUATION

Laboratory testing has a limited role in the evaluation of the patient with hip pain. A complete blood count, erythrocyte sedimentation rate, and C-reactive protein may be useful in patients with suspected infection. For patients with possible inflammatory arthritis or spondyloarthropathies, rheumatoid factor and human leukocyte antigen B27 should be included in the blood work.

RADIOGRAPHIC EVALUATION

Plain radiographs are essential in the evaluation of the patient with hip pain and aid in the diagnosis of most intraarticular hip disorders. An anteroposterior (AP) view of the pelvis and involved hip and a frog-leg lateral view provide information about leg lengths, degenerative or dysplastic changes, and bony lesions in the pelvis or proximal femur. When appropriate, an AP radiograph taken with the patient standing and a false profile view can help to visualize more subtle joint abnormalities. Ultrasonography is useful for diagnosing hip effusion, particularly in the pediatric population. A bone scan may be useful in diagnosing degenerative conditions and tumors, but is less helpful in diagnosing chondral defects and labral tears. A computed tomography (CT) scan may detect bony abnormalities but is less effective for evaluating soft tissue. Magnetic resonance imaging (MRI) of the hip may be used to assess soft tissue abnormalities and subtle degenerative changes. Magnetic resonance imaging also is useful in staging avascular necrosis, but is not accurate in evaluating labral or chondral injury.¹⁴ The use of gadolinium as a contrast agent with MRI improves detection of labral disorders, but there still is a high false-negative rate.^{3,6,10,16}

HIP ARTHROSCOPY

In some patients, the diagnosis remains elusive despite a complete history, physical examination, and appropriate radiographic and laboratory testing. These patients may benefit from diagnostic hip arthroscopy, particularly if there is a suggestion of labral or chondral disorders or loose bodies during the initial evaluation.

DIFFERENTIAL DIAGNOSIS

Approximately 2.5% of all sports-related injuries are located in the hip area, and this number increases to 5% to 9% in the high school athlete.¹ Runners and soccer players tend to be more prone to injuries of the hip and groin than

athletes who participate in other sports. Traumatic and overuse soft tissue injuries include muscular, bursal, tendinous, or ligamentous inflammation, contusions, strains, and sprains. Skeletal injuries can involve the physis or apophysis in children, and skeletal disorders includes fractures, subluxations, dislocations, stress injuries, infections, and avulsions. Patients with nontraumatic hip pain from systemic conditions such as rheumatoid arthritis, juvenile arthritis, ankylosing spondylitis, tumors, and metabolic bone disease also may present to the sports physician. These conditions should be suspected when the severity or course of the injury is atypical.

Persistent hip pain can originate from intraarticular disorders such as avascular necrosis, osteoarthritis, loose bodies, labral tears, or pyarthrosis. Hip pain also may be secondary to a lumbar spine disorder. Nerve entrapment syndromes involving the ilioinguinal, genitofemoral, and lateral femoral cutaneous nerve of the thigh may present as hip pain or paresthesias.

Contusions

Contusions result from a direct blow to a specific soft tissue area, typically over a bony prominence in the pelvis or the greater trochanter. Contusions are most common in athletes who participate in contact sports such as football and hockey, but also are seen from contact with equipment in sports such as gymnastics and from contact with high velocity balls in sports such as lacrosse. Point tenderness, ecchymosis, and hematoma are common findings with this type of injury. It is important to determine the extent of hemorrhage and hematoma formation because this will impact when an athlete is ready for return to play. Radiographs may show an avulsion fracture associated with the injury.

Strains

The majority of soft tissue injuries about the hip and pelvis are musculotendinous strains. A strain is an injury to a musculotendinous structure caused by an indirectly applied force,

typically with eccentric contraction or overstretching of an activated muscle. Low rates of loading result in failure at the tendon-bone junction with avulsion or disruption at the tendinous insertion. High rates of loading typically cause a midsubstance tear in the musculotendinous unit. This type of injury varies depending on the patient's age, physical condition, and the mechanism of injury. Causes range from repetitive microtrauma to a discrete macrotraumatic event. Strains are reported on a three-scale clinical grading system. Grade I injuries are a simple stretching of the musculotendinous unit. Grade II injuries involve partial tearing, and Grade III injuries result in complete disruption.²

Athletic Pubalgia

Athletic pubalgia is a chronic inguinal or pubic pain in athletes that worsens with exertion and is not attributable to an occult hernia.^{7,15,17} The rectus tendon insertion into the pubis, the external oblique muscle and aponeurosis, and the adductor longus are thought to be the primary sites of the pain. In some patients, the pain follows a discrete hyperextension injury but in other patients the onset is insidious. The pain often is severe enough to limit activity. Magnetic resonance imaging findings are nonspecific in the majority of patients, and surgery often is required for definitive resolution.

Piriformis Syndrome

Piriformis syndrome is a compression of the sciatic nerve as it exits the pelvis under the piriformis muscle.¹⁵ Patients have pain in a sciatic distribution. Activities that require hip flexion and internal rotation typically are painful, and patients may have tenderness over the piriformis tendon. Pace's sign, pain with resisted abduction and external rotation of the thigh, is positive.¹⁵ An MRI scan often will show sciatic nerve inflammation in the area of the piriformis.

Hamstring Syndrome

Hamstring syndrome is a pain radiating from the ischial tuberosity down the posterior aspect of the thigh into the popliteal fossa.¹⁵ The pain is exacerbated with hamstring stretch in

activities such as sprinting and hurdling. Physical examination reveals tenderness over the ischial tuberosity and pain with resisted leg extension. A possible etiology is entrapment of the sciatic nerve between the semitendinosis and biceps femoris muscles.

Inflammatory Disorders

Many of the spondyloarthropathies, including Reiter's syndrome, psoriatic arthritis, ankylosing spondylitis, and inflammatory bowel disease can result in inflammation of the sacroiliac joint. Pain occurs over the sacroiliac joint or is referred to the lower buttock and thigh. The patient has pain with compression of the pelvis or hyperextension of the hip. Radiographs may show blurring of the joint in advanced cases. Rheumatoid arthritis commonly results in inflammatory changes in the hip.

Snapping Hip Syndrome

Snapping hip syndrome is a collection of extraarticular and intraarticular disorders that are painful, disabling, and difficult to diagnose. The patient complains of pain and popping in the hip, and occasionally has a sensation of dislocation.

Extraarticular sources include the iliopsoas tendon over the iliopectineal prominence or the lesser trochanter, the iliofemoral ligaments over the femoral head, the long head of the biceps over the ischial tuberosity, or most commonly the iliotibial band over the greater trochanter. Intraarticular causes include labral tears, synovitis, and loose bodies. Currently, diagnosis is difficult but can be aided by CT, MRI, and bursography and occasionally by hip arthroscopy (Fig 1).

Bursitis

Bursitis is a common cause of hip pain related to inflammation of one of the three main bursae of the hip: the trochanteric bursa, the iliopsoas bursa, or the ischiogluteal bursa. Bursitis may be caused by overuse or degenerative changes in the bursae. Patients with trochanteric bursitis present with pain over the greater trochanter that is exacerbated by hip adduction. Ischiogluteal

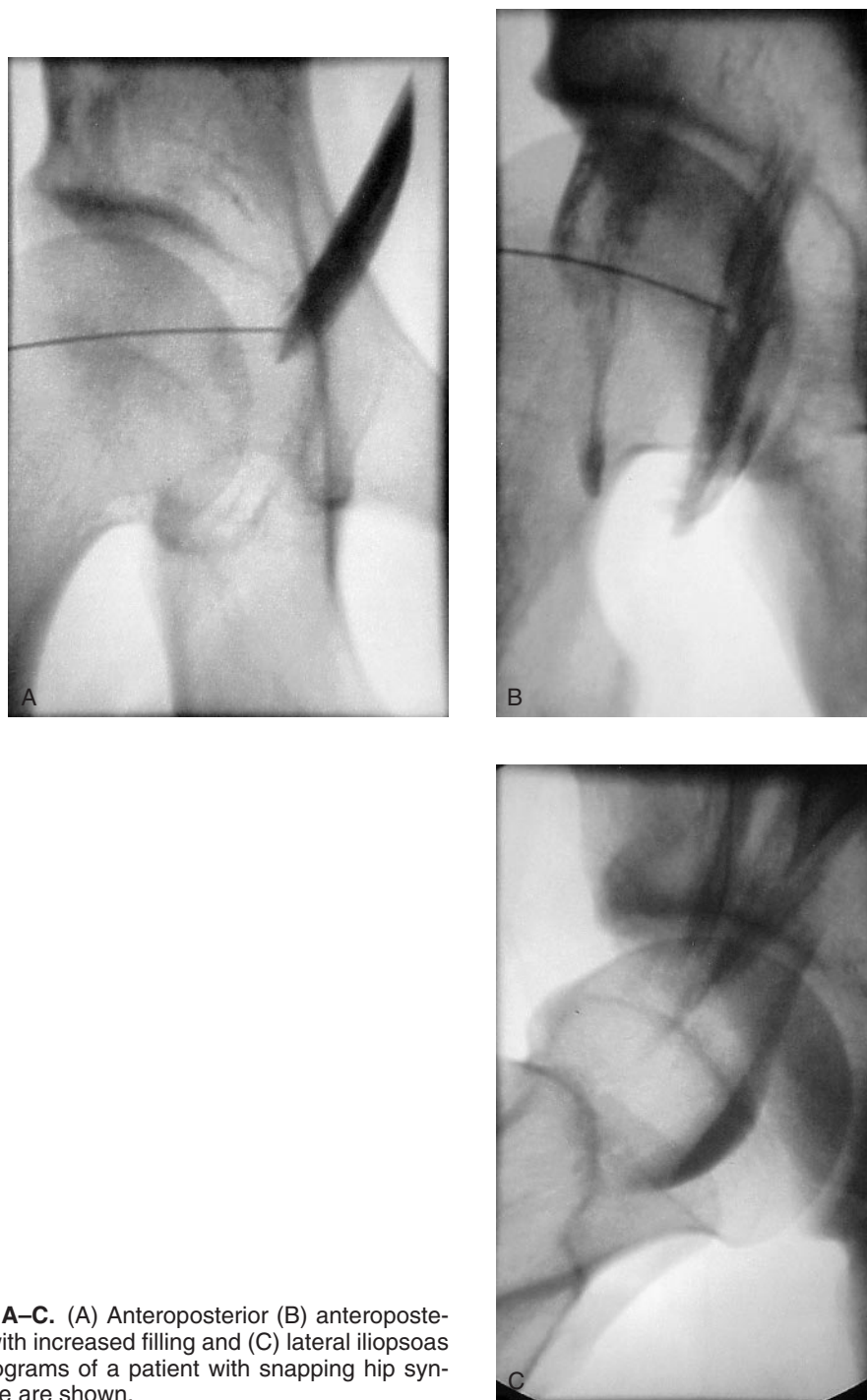


Fig 1A–C. (A) Anteroposterior (B) anteroposterior with increased filling and (C) lateral iliopsoas bursograms of a patient with snapping hip syndrome are shown.

bursitis often is associated with sitting for long periods. Radiographs typically are not helpful unless there are chronic calcific changes.

Arthritis

Osteoarthritis is a common condition affecting the hip in adults. Osteoarthritis may be the result of a pediatric hip problem or past hip trauma, but often has no identifiable cause. Motion in the hip becomes progressively restricted because of synovitis, soft tissue contractures, and loss of joint congruency. The pain in osteoarthritis is attributable to synovitis, muscle spasm, and pain fibers within exposed bone and granulation tissue. Patients complain of pain in the groin, buttock, anterior thigh, or knee, and often have an antalgic gait. Pain is worse with weightbearing but may be present at rest and increase progressively. Examination of the hip shows limited ROM and often shows a flexion contracture. Abduction and internal rotation are most restricted. Radiographs reveal joint space narrowing, cyst and osteophyte formation, and sclerotic changes in the subchondral bone with superolateral subluxation of the femoral head.

Septic Arthritis

Pyarthrosis is rare in the hip in adults. Patients who are immunocompromised, patients with renal failure or diabetes mellitus, patients who are intravenous drug abusers, and patients receiving corticosteroids or chemotherapeutic agents are at increased risk for septic arthritis. Patients with a septic hip present with high fever, exquisite pain, and decreased motion. Radiographs usually are normal acutely but may show a widened joint space secondary to effusion.

Osteonecrosis

Osteonecrosis of the femoral head is well-recognized in association with femoral neck fractures and hip dislocations (1%–17%) in children and adults.¹¹ Osteonecrosis as a result of trauma is caused by disruption of the arterial supply to the femoral head, and also may be associated with showers of fat em-

boli. Risk factors for osteonecrosis in non-traumatic circumstances include corticosteroid use, alcohol abuse, sickle-cell disease, gout, Gaucher's disease, caisson disease, and hypercoagulable states. However, in many patients the cause is idiopathic. The cause of osteonecrosis in situations without recognizable injury is not well-understood but may be caused by vascular thrombosis, venous compression, or fat embolism.

The pathologic change associated with the early phase of the disease is segmental necrosis of the femoral head. The overlying cartilage, nourished by synovial fluid, is unaffected. With time, the necrotic bone is resorbed, and subchondral collapse may occur, resulting in joint incongruency and pain.

Advanced stages of osteonecrosis are visible on plain radiographs, and MRI is useful in early stages of the disease. Osteonecrosis often is bilateral in nontraumatic cases.

Labral Tears

The torn acetabular labrum has been identified as a cause of hip pain in athletes. Clinical features include a painful click in the inguinal area that radiates toward the gluteus, catching, and occasionally giving way. Athletes often remember a traumatic event involving hip flexion, abduction, and forceful knee extension. The painful click may be reproducible by a McCarthy sign (after both hips are flexed fully, the affected hip is extended, first in external rotation, then in internal rotation) (Fig 2). Radiographs and MRI scans have a low diagnostic yield. In patients with refractory hip pain, reproducible physical findings, and equivocal or negative radiographic studies, hip arthroscopy is useful for diagnosis and treatment.

Fractures and Dislocations

Traumatic causes of fractures and dislocations are beyond the scope of this paper. A stress fracture is defined as repetitive stress below the failure level of bone in a period insufficient to allow for bony remodeling. Stress fractures of the femur and pelvis are uncommon but often are misdiagnosed. These injuries should be

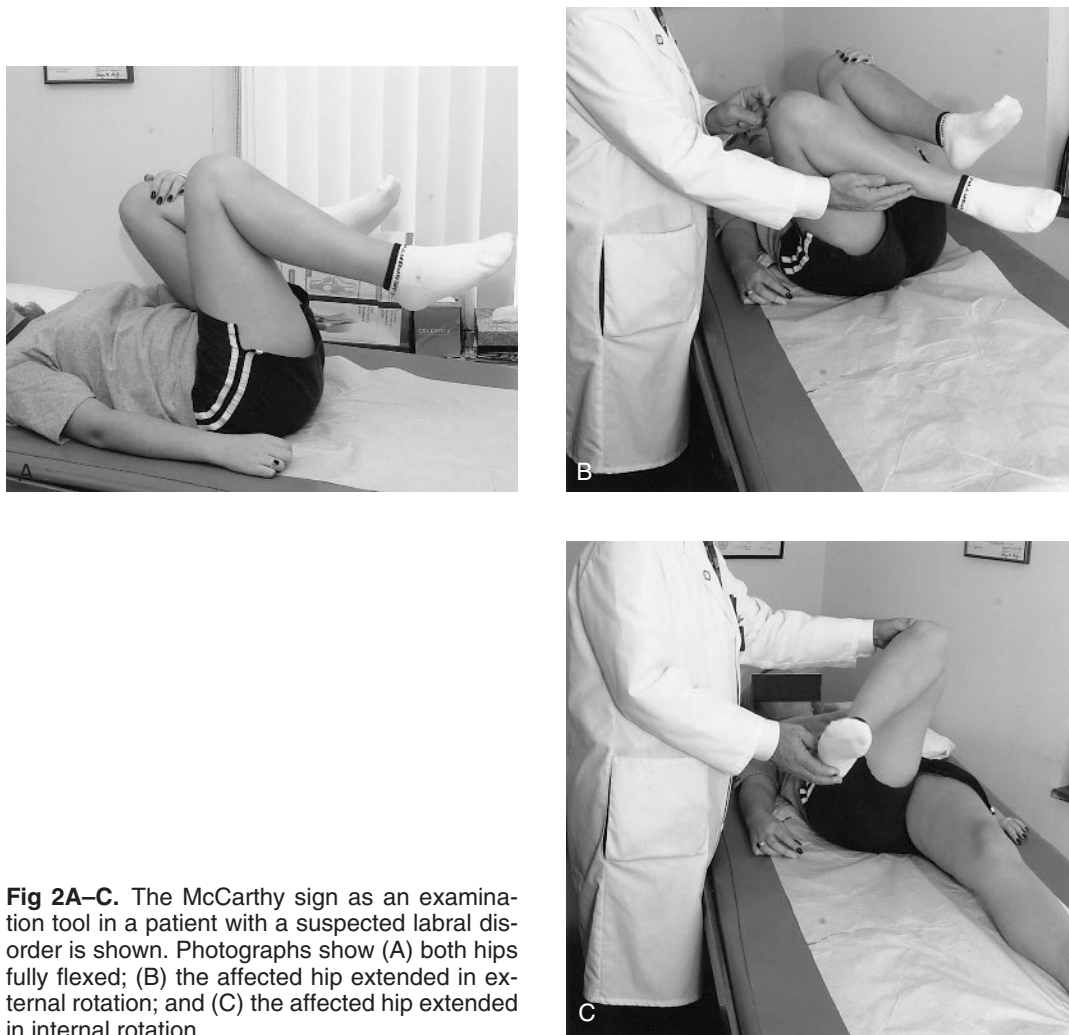


Fig 2A–C. The McCarthy sign as an examination tool in a patient with a suspected labral disorder is shown. Photographs show (A) both hips fully flexed; (B) the affected hip extended in external rotation; and (C) the affected hip extended in internal rotation.

considered in patients who participate in activities involving chronic repetitive motion such as track and field, long distance running, and military marching.

Femoral stress fractures are less common than those involving the pelvis but may lead to fracture displacement and osteonecrosis if not treated appropriately. Athletes present with persistent groin pain and complain of decreased ROM, which is confirmed by physical examination. Initial plain radiographs may be negative for the first 2 to 4 weeks, but bone scans and MRI scans are helpful in confirming the fracture.

Two types of femoral neck stress fractures have been described: distraction and compression.¹⁹ Distraction stress fractures usually are transverse and involve the superior portion of the neck. They are prone to displacement and are uncommon in children. Compression type fractures are seen along the inferior medial aspect of the femoral neck and typically do not displace. They are seen more commonly in children.

Pelvic stress fractures also are secondary to repetitive microtrauma and occur at the junction of the ischium and inferior pubic ramus. They are common in female distance runners.

Athletes present with inguinal pain, an antalgic gait, and full ROM. Plain radiographs may be negative for as many as 2 to 3 weeks after injury.

Tumors

Hip or pelvic pain that persists for longer than expected after an apparently minor injury or without a traumatic event should be evaluated for a possible pathologic lesion, especially in older patients.

The chief complaint of children with a pathologic lesion in the hip or pelvis is groin pain, although quality, location, and the nature of the pain varies. Most pediatric patients with musculoskeletal tumors do not have associated systemic symptoms. A thorough examination of the pelvis and hip will determine the presence of masses, muscular atrophy, or neurovascular changes. Plain radiographs often are diagnostic, but bone scans, CT scans, or MRI scans may be useful in selected patients.

Osteoid osteoma is a benign bone tumor of adolescents and young adults frequently involving the hip and pelvis. Plain radiographs may be normal but tomograms, bone scans, or CT scans will identify the classic bony nidus. Other common lesions about the hip include unicameral bone cysts, osteochondroma, fibrous dysplasia, osteogenic sarcoma, Ewing's sarcoma, and brown's tumor.

Hip pain is a common problem with many etiologies that is encountered by orthopaedists. The source of the pain usually is evident after a thorough history and physical examination and appropriate laboratory and radiographic evaluation. Hip arthroscopy may be helpful if the diagnosis remains elusive.

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